Linqiang (Jackson) Guo

AI PhD Researcher · Software Engineering · Concordia University & Mila
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RESEARCH INTERESTS

LLM/VLM-based autonomous agents; mobile GUI understanding and automation; multimodal VLM post-training (SFT, DPO, GRPO, RLHF); state-aware and recoverable agent reasoning; robust automated GUI testing; AI for software engineering.

EDUCATION

Ph.D., Software Engineering — Concordia University 2024 – Present
Montreal, Canada

M.S., Electrical & Computer Engineering (AI track) — University of Miami 2020 – 2022
Miami, USA

B.S., Electrical Engineering — University of Missouri Columbia 2016 – 2019
Columbia, USA

EXPERIENCE

Graduate Student Researcher — Mila (Quebec AI Institute) Jul 2026 – Present
Montreal, Canada

- Research on multimodal / GUI agents.
- RL-based post-training for agentic models (GRPO, DPO).

Research Assistant — SPEAR Lab, Concordia University Jan 2024 – Present
Montreal, Canada

- Research intelligent mobile software systems focused on AI-driven automation and UI understanding.
- Develop methods to improve agent reasoning, state awareness, and robustness in complex mobile environments.
- Co-authored papers at AAAI, ICSE, ECCV (workshop), and COLM (workshop).

Teaching Assistant — Concordia University Jan 2025 – Jan 2026
Montreal, Canada

- Assisted teaching Software Engineering: software design, UML modeling, testing, and project management.
- Led weekly tutorials and supported students through office hours and online discussions.

Software Developer — Walmart Global Tech Aug 2022 – Dec 2023
Sunnyvale, California, USA · in contract

- Built and maintained microservices for customer personalization, data ingestion, and recommendation using Java, Spring, and cloud tools.
- Optimized the Milvus vector database on Kubernetes; integrated GenAI workflows into search, improving query understanding and relevance ranking.
- Deployed scalable API endpoints and streamlined CI/CD with data science and platform teams.

Software Engineer Intern — CPECC, Gulf Region Jan 2020 – Feb 2020
Abu Dhabi, UAE

- Developed RESTful web services with Spring Boot, Spring MVC, Maven, and Hibernate; tested with JUnit and Mockito.

PUBLICATIONS

[1] Learning to Adapt: Few-Shot Generalization of Mobile Agents to Unseen Applications
Siobhan Reid; Li Gu; Zhixiang Chi; **Linqiang Guo**; Zihuan Jiang; Ziqiang Wang; Yuanhao Yu; Yang Wang
Best Paper Award. ECCV 2026 Workshop on Multimodal Digital Agents (MDA), Spotlight Talk + Poster.

[2] Q-Reflector: Learning GUI Action Reflection with Generated Conditions
Linqiang Guo; Wei Liu; Li Gu; Yang Wang; Tse-Hsun (Peter) Chen
ECCV 2026 Workshop on Multimodal Digital Agents (MDA), Poster + Lightning Talk. arXiv:2608.05587

[3] CoAdapt-GUI: Joint Workflow Context and Policy Adaptation for Unseen GUI Applications
Linqiang Guo; Li Gu; Zihuan Jiang; Zhixiang Chi; Siobhan Reid; Ziqiang Wang; Yuanhao Yu; Wei Liu; Yang Wang; Tse-Hsun (Peter) Chen
ECCV 2026 Workshop on Multimodal Digital Agents (MDA), Poster + Lightning Talk. arXiv:2608.11588

[4] Beyond the GUI Paradigm: Do Mobile Agents Need the Phone Screen?
Li Gu; Zihuan Jiang; **Linqiang Guo**; Zhixiang Chi; Ziqiang Wang; Huan Liu; Yuanhao Yu; Tse-Hsun (Peter) Chen; Yang Wang
ECCV 2026 Workshop on Multimodal Digital Agents (MDA), Poster. arXiv:2606.19388

[5] AgentPersonaBench: Benchmarking Persona-Driven User Simulation
Jintao Huang; Yifan Wang; Hongyu Shen; ...; **Linqiang Guo**; ...; Xiaomin Li; Yuexing Hao (46 authors)
Preprint, 2026. arXiv:2610.04379

[6] Agent-SAMA: State-Aware Mobile Assistant
Linqiang Guo; Wei Liu; Yi Wen Heng; Yang Wang; Tse-Hsun (Peter) Chen
AAAI 2026. arXiv:2505.23596

[7] GUIWatcher: Automatically Detecting GUI Lags by Analyzing Mobile Application Screencasts
Wei Liu; Feng Lin; **Linqiang Guo**; Tse-Hsun (Peter) Chen; Ahmed E. Hassan
ICSE 2025. arXiv:2502.04202

[8] MAPLE: A Mobile Assistant with Persistent Finite State Machines for Recovery Reasoning
Linqiang Guo; Wei Liu; Yi Wen Heng; Yang Wang; Tse-Hsun (Peter) Chen
COLM 2025 Workshop.

[9] Registration on DCE-MRI Images via Multi-Domain Image-to-Image Translation
Naxin Cai; Houjin Chen; Yanfeng Li; Yahui Peng; **Linqiang Guo**
Computerized Medical Imaging and Graphics (IF 7.422).

[10] Screencast-Based Analysis of User-Perceived GUI Responsiveness
Wei Liu; **Linqiang Guo**; Yi Wen Heng; Chenglin Li; Tse-Hsun (Peter) Chen; Ahmed E. Hassan
Under review. arXiv:2508.01337

[11] PopSweeper: Automatically Detecting and Resolving App-Blocking Pop-Ups to Assist Automated Mobile GUI Testing
Linqiang Guo; Wei Liu; Yi Wen Heng; Yang Wang; Tse-Hsun (Peter) Chen
Under review.

HONORS & AWARDS

Best Paper Award — ECCV 2026 Workshop on Multimodal Digital Agents (MDA) 2026
Concordia University International Tuition Award of Excellence 2023
Concordia University Split Graduate Fellowship 2023

SKILLS

Languages: Python, Java, C, SQL, HTML/CSS, MATLAB

AI / ML: Multimodal VLM post-training (SFT, DPO, GRPO, RLHF), LLM agents, PyTorch, HuggingFace, Unsloth, OpenCV

Systems & Tools: Spring Boot, Spring Cloud, Kubernetes, AWS, MySQL, Linux, ADB, Git